

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | M.Sc. Candidate in Applied AI

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To: Selection Committee

Digital, AI and Innovation (DAI) Hub | Bureau for Policy and Programme Support (BPPS)

United Nations Development Programme (UNDP)

Job Identification: 33001 — Digital, AI and Innovation Internship: Global Call for 2026

SUBJECT: Application for Digital, AI and Innovation Internship (Home-Based)

Dear Members of the Selection Committee,

I am writing to express my enthusiastic application for the Digital, AI and Innovation Internship within UNDP's Bureau for Policy and Programme Support (BPPS). Currently enrolled in a Master's degree in Applied Artificial Intelligence at the University of Ngaoundéré (Cameroon) following a Civil Engineering degree (B.Sc.), I design sovereign, ethical, and high-impact AI systems. I am eager to contribute my technical acumen, product development experience, and commitment to digital public goods to UNDP's mission of accelerating the Sustainable Development Goals (SDGs).

🔗 **AI Product Build & Systems Innovation:** As founder of Archi Cam AI (official applicant to the Google Africa Applied AI Lab 2026), I developed an agentic platform combining Gemma and Gemini LLMs with 5D BIM standards to automate construction cost estimation and planning ($<45s$, $R^2=0.9872$). Furthermore, I engineered K1-MATHINFO v3, a sovereign multi-agent GraphRAG architecture indexing 470 academic theses with Neo4j and LangGraph, incorporating OKF zero-hallucination verification protocols.

🔗 **Data-Driven Impact for Climate & Vulnerable Regions:** Driven by the 'Leave No One Behind' principle, I co-developed VigieSahel, an agro-climatic forecasting system combining XGBoost predictive models and Supabase to reduce crop seeding losses by 35% and anticipate epidemic risks by 14 days in the Sahel region. Additionally, through Dataset Automator (Google Cloud Hackathon), I built MLOps pipelines implementing KS/PSI data drift monitoring and EU AI Act alignment frameworks.

🔗 **Rigor, Ethics & Operational Crisis Management:** In addition to my AI expertise, my background as an Aviation Security Officer (AVSEC) at the Cameroon Civil Aviation Authority has instilled a deep respect for strict international regulatory frameworks (ICAO Annex 17), crisis coordination, and high-pressure delivery. I apply this operational discipline to AI governance, emphasizing explainability (SHAP Sentinel Audit), fairness, and responsible data policies.

Joining the UNDP DAI Hub represents an exceptional opportunity to put cutting-edge technology at the service of Country Offices, local communities, and multilateral digital transformation. Available for a 3-month home-based arrangement, I am prepared to contribute immediately with dedication, cultural adaptability, and high energy.

Thank you very much for your time and consideration. I welcome the opportunity to discuss my application with you.

Sincerely,

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Graduate Student in Applied AI | Founder @ Archi Cam AI